Home Lab — Technical Documentation

Enterprise IT Infrastructure Build — Complete Reference

James Mitchell | March 2026 | Version 2.0

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1. Project Overview

This document provides complete technical documentation for the Wizardwerks Enterprise Solutions home lab — a multi-server Windows enterprise environment built from scratch to simulate a real-world small business IT infrastructure. The lab was built to demonstrate hands-on enterprise IT administration skills including Active Directory, DNS, DHCP, Group Policy, file services, PowerShell automation, help desk ticketing, patch management, backup and disaster recovery, and network security via pfSense.

Lab Name	Wizardwerks Enterprise Solutions
Domain Name	CORP.local
Platform	Oracle VM VirtualBox
Host OS	Windows (physical host machine)
Host Storage Available	161 GB free
Build Start	March 2026
Documentation Version	2.0
Administrator	James Mitchell

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2. Host Machine

The physical host machine runs VirtualBox and hosts all virtual machines. It serves as the hypervisor layer for the entire lab environment.

Role	Hypervisor — VirtualBox Host
Virtualization Platform	Oracle VM VirtualBox
Free Storage	161 GB available for backup repositories and VM expansion
Network Adapters on VMs	Internal Network (VM-to-VM) + NAT (internet access)
Drag and Drop	Bidirectional — enabled for file transfers between host and VMs
Backup Repository	E: drive on FS01 — dedicated 30GB backup volume

3. Virtual Machines

3.1 — DC01 (Domain Controller)

DC01 is the primary domain controller and the backbone of the entire lab environment. It hosts all core identity and network services.

Machine Name	DC01
Role	Primary Domain Controller
Operating System	Windows Server 2022
Domain	CORP.local
Storage Used	10.39 GB
Storage Available	39 GB
Network Adapter 1	Internal Network — VM-to-VM communication
Network Adapter 2	NAT — internet access for updates and downloads
vCPU	1 vCPU (single core — lab environment)

Services Running on DC01

- Active Directory Domain Services (AD DS)
- DNS Server — authoritative for CORP.local
- DHCP Server — IP address assignment for domain devices
- Group Policy Management
- WSUS (Windows Server Update Services) — patch management, GPO deployed
- Spiceworks Help Desk Agent — cloud help desk integration

Active Directory Configuration

- Forest/Domain: CORP.local
- Organizational Units: HR, IT, Sales
- Security Groups: HR_Users, IT_Users, Sales_Users — one per OU
- Each OU contains test user accounts provisioned via PowerShell
- Group Policy Objects linked to each OU
- Drive mapping GPO configured and verified with gpresult /r

DNS Configuration

- Primary DNS zone: CORP.local — Active Directory integrated
- DC01 is the authoritative DNS server for the internal domain
- All domain devices point to DC01 for DNS resolution
- External DNS forwarding configured for internet name resolution

DHCP Configuration

- DHCP scope configured to assign IPs within defined range
- DC01 serves as DHCP server for all lab VMs
- Lease duration: standard default
- APIPA fallback (169.254.x.x) confirmed as expected behavior when DHCP unavailable

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3.2 — FS01 (File Server)

FS01 is the dedicated file server hosting department shared drives, home directories, and the backup repository. Windows Server Backup runs scheduled nightly jobs to the dedicated E: backup volume.

Machine Name	FS01
Role	File Server / Backup Server
Operating System	Windows Server 2022
Domain	CORP.local — domain joined
Network Adapter 1	Internal Network — communication with DC01
Network Adapter 2	NAT — internet access confirmed
vCPU	1 vCPU — acceptable for lab environment
Internet Connectivity	Verified — Test-NetConnection google.com = True

File Server Configuration

- Share root: C:\Shares\HomeDirectories
- UNC path: \\FS01\HomeDirectories
- Share permissions: Domain Admins — Full Control, Domain Users — Change
- NTFS permissions: configured per department security group
- Department shares: HR, IT, Sales — each accessible only to corresponding AD security group
- Home directories: created automatically per user via PowerShell provisioning script
- Home directory path per user: \\FS01\HomeDirectories\[username]
- H: drive mapped in AD user profile properties

NTFS Permission Structure

\\FS01\HomeDirectories\HR	HR_Users — Full Control (own folder only)
\\FS01\HomeDirectories\IT	IT_Users — Full Control (own folder only)
\\FS01\HomeDirectories\Sales	Sales_Users — Full Control (own folder only)
All department folders	Domain Admins — Full Control

3.3 — Workstation VM

A Windows workstation VM simulates an end user machine within the domain. Used for testing Group Policy application, user login behavior, drive mapping, and troubleshooting scenarios from the client perspective.

Role	End User Workstation
Domain	CORP.local — domain joined
Purpose	Client simulation, GPO testing, troubleshooting practice
Network Adapter	Internal Network — communicates with DC01 and FS01

3.4 — pfSense VM (Network Firewall/Router)

pfSense is an open source firewall and router platform running as a dedicated VM. It provides network segmentation, firewall rule management, VLAN capability, and inter-network routing for the Wizardwerks lab environment.

Machine Name	pfSense
Role	Firewall / Router / Network Security
Platform	pfSense CE (Community Edition)
OS Base	FreeBSD
Storage	8 GB — dynamically allocated
Network Adapter 1 (WAN)	NAT — internet facing side
Network Adapter 2 (LAN)	Internal Network — connects to lab environment
Status	Installed — initial configuration in progress

pfSense Planned Configuration

- VLAN segmentation — separate network zones for different lab functions
- Firewall rules — control traffic between VLANs and to/from internet
- Inter-VLAN routing — managed and controlled through pfSense
- DHCP relay or dedicated DHCP per VLAN
- Traffic monitoring and logging
- Web interface accessible from internal network for configuration

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4. PowerShell Automation

Two PowerShell provisioning scripts were developed and are stored at C:\Scripts\ on DC01. Both scripts automate the full user onboarding workflow including AD account creation, OU placement, security group assignment, home folder creation, NTFS permissions, and H: drive mapping.

4.1 — Single User Provisioning Script

- Interactive Read-Host prompts for all user details
- Yellow confirmation block displays all entered information before proceeding
- Y/N gate — requires explicit confirmation before any changes are made
- Try/Catch error handling throughout
- $CurrentStep variable tracks exactly which step failed if an error occurs
- AD user account creation with full attribute population
- OU placement based on department input
- Security group assignment — adds user to department security group
- Home folder creation at \\FS01\HomeDirectories\[username]
- NTFS permissions set — Full Control for user on their own folder
- H: drive mapping configured in AD user profile
- Cyan success summary displayed on completion

📝 *Script Location: C:\Scripts\NewUser.ps1*

4.2 — Bulk User Provisioning Script (CSV)

- Reads from C:\Scripts\NewUsers.csv
- ForEach loop processes all users in the CSV sequentially
- Same full provisioning workflow as single user script applied per user
- Real-time per-user SUCCESS / FAILED reporting during execution
- Final summary count displayed on completion — Success: X, Failed: X
- Tested with Justice League CSV — 6 users, 6/6 success

📝 *Script Location: C:\Scripts\BulkProvision.ps1*

📝 *CSV Format: FirstName, LastName, Department, Username, Password*

📝 *Test dataset: Batman, Superman, Wonder Woman, Flash, Aquaman, Green Lantern — 6/6 SUCCESS*

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5. Group Policy Configuration

5.1 — Drive Mapping GPO

- Linked to: All department OUs — HR, IT, Sales
- Function: automatically maps H: drive to user home directory at login
- Path mapped: \\FS01\HomeDirectories\%username%
- Verified with: gpresult /r — confirmed GPO appearing in Applied Group Policy Objects

5.2 — Key GPO Management Commands

- `gpresult /r` — show all GPOs applying to current user/computer
- `gpupdate /force` — force immediate Group Policy refresh
- `Get-GPO -All` — list all GPOs in PowerShell
- `gpresult /h C:\report.html` — generate HTML Group Policy report

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6. Spiceworks Help Desk

6.1 — Configuration

- Platform: Spiceworks Cloud — browser-based, no local server required
- Agent: installed on DC01 for asset and network visibility
- Access: via web browser from any machine on the network
- Account: registered and active

6.2 — Completed Tickets

Ticket #1	John Doe — Unable to access shared drive — Network
Ticket #2	Jane Smith — New user account request — Other
Ticket #3	Tony Stark — Workstation running slow — Hardware
Ticket #4	Bruce Wayne — Password reset — Other

📝 *All 4 tickets completed full lifecycle: Create > Assign > Internal Note > Public Response > Close*

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7. Backup & Disaster Recovery

Windows Server Backup is the backup and disaster recovery solution implemented on FS01. It is a native Windows Server role — no additional software required. Backup jobs are scheduled nightly, and restore capability has been verified through a successful test recovery.

7.1 — Configuration

- Role installed via Server Manager > Add Roles and Features > Windows Server Backup
- Dedicated backup volume: E: drive — 30GB VDI dynamically allocated, attached to FS01
- E: drive initialized as MBR, formatted NTFS, labeled Backup
- Backup schedule: nightly at 9:00 PM
- Backup type: Full Server
- Backup destination: E: (Backup) volume

7.2 — Restore Verification

- Test restore completed successfully via Windows Server Backup > Recover
- Recovery type: Files and folders
- Restore option: Create copies — preserves existing files
- Result: Completed successfully — backup and restore capability verified

📝 *CRITICAL: A backup without a tested restore is only half the story. Always verify. This has been done.*

7.3 — Key Commands

- wbadmin get versions — list available backup versions
- wbadmin start backup — initiate a manual backup job
- wbadmin start recovery — initiate a manual restore
- Windows Server Backup GUI: Server Manager > Tools > Windows Server Backup

7.4 — Tool Selection Rationale

Veeam Backup and Replication was evaluated but determined to be overspecced for the current lab environment due to resource requirements and service dependencies. Windows Server Backup was selected as a native, lightweight, production-grade alternative. This decision reflects real-world judgment — matching the right tool to the environment rather than defaulting to the most complex option.

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8. WSUS — Patch Management

Windows Server Update Services (WSUS) provides centralized patch management for the Wizardwerks domain. Installed as a built-in Windows Server role on DC01 — no additional download required. All domain computers receive updates from DC01 rather than pulling directly from Microsoft.

8.1 — Installation

- Installed via Server Manager > Add Roles and Features > Windows Server Update Services on DC01
- Role services: WSUS Services and WID Connectivity
- Content directory: C:\WSUS
- Post-installation tasks completed via Server Manager notification

8.2 — Configuration

- Upstream server: Microsoft Update — DC01 pulls directly from Microsoft
- Products synced: Windows Server 2019 and Later (covers Server 2022), Windows 10
- Classifications: Critical Updates, Security Updates, Update Rollups
- Sync schedule: daily automatic synchronization
- Initial synchronization completed successfully

8.3 — Computer Groups

- Domain Controllers — DC01
- Servers — FS01
- Workstations — Windows 10 workstation VM

8.4 — Group Policy Configuration

- GPO name: WSUS Policy — created and linked to CORP.local domain
- Specify intranet Microsoft update service location: http://DC01:8530 (both fields)
- Configure Automatic Updates: 4 - Auto download and schedule the install
- Scheduled install: daily at 3:00 AM
- GPO enforced and applied via gpupdate /force — verified

8.5 — Key Concepts

- Centralized patch management — no individual machines pulling directly from Microsoft
- Staged rollout capability — test group first, then broader deployment
- WSUS + Windows Server Backup = complete patch management and DR story
- Always run a backup before patch deployment — rollback capability if something breaks

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9. BitLocker Drive Encryption

BitLocker drive encryption was implemented on DC01 to protect data at rest. BitLocker is a native Windows Server feature that encrypts the entire drive — making data unreadable without the correct authentication. This is a compliance best practice for any environment handling sensitive data.

9.1 — Implementation Approach

Virtual machines in VirtualBox do not have a physical TPM (Trusted Platform Module) chip by default. Rather than enabling UEFI on an existing VM — which caused boot failure during testing — a Group Policy workaround was used to configure BitLocker without TPM. This is a supported and documented enterprise configuration.

- BitLocker feature installed via PowerShell: Install-WindowsFeature BitLocker -IncludeAllSubFeature -IncludeManagementTools
- GPO created: BitLocker Policy — linked to CORP.local domain
- Policy enabled: Require additional authentication at startup > Allow BitLocker without a compatible TPM
- GPO applied via gpupdate /force
- Access method: password-based unlock at startup

9.2 — DC01 BitLocker Configuration

- Status: ENABLED — C: drive fully encrypted
- Encryption mode: XTS-AES (New Encryption Mode — strongest available)
- Unlock method: Password
- Scope: Used disk space only — faster encryption, sufficient for lab environment
- Recovery key: Saved securely
- Verified: Drive locked on restart, password prompted, system boots correctly after authentication

📝 *STATUS: COMPLETE — DC01 C: drive encrypted and verified*

9.3 — FS01 BitLocker

- Status: Deferred — Guest Additions not installed on FS01, preventing shared folder access needed to save recovery key
- DC01 encryption is sufficient for lab demonstration purposes
- FS01 BitLocker can be completed when Guest Additions are installed

📝 *STATUS: DEFERRED — not blocking lab completion*

9.4 — Key Commands

- manage-bde -status C: — check BitLocker status and encryption percentage
- manage-bde -on C: — enable BitLocker on C: drive
- manage-bde -unlock C: — unlock drive with recovery key
- Get-BitLockerVolume — PowerShell status check

9.5 — Why This Matters

Encryption at rest is a baseline security requirement for any environment handling sensitive data. Financial institutions (NCUA), healthcare (HIPAA), and legal environments all require drive encryption as part of their compliance posture. Being able to implement BitLocker — including the GPO workaround for non-TPM environments — is a directly sellable and demonstrable skill.

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10. Network Architecture

9.1 — Current Topology

- Host machine runs VirtualBox hypervisor
- DC01, FS01, Workstation, and pfSense communicate via Internal Network adapter
- All VMs have NAT adapter for internet access
- DC01 provides DNS and DHCP services to the internal network
- FS01 and Workstation joined to CORP.local domain — authenticate via DC01
- pfSense acts as firewall and router between network segments

9.2 — Connectivity Verification Commands

- `Test-NetConnection -ComputerName google.com -Port 80  # Test internet`
- `Test-NetConnection -ComputerName DC01 -Port 389  # Test AD connectivity`
- `Test-NetConnection -ComputerName FS01 -Port 445  # Test file share`
- `ping DC01  # Basic ping test`
- `nslookup CORP.local  # Test DNS resolution`

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11. Key Commands Reference

Active Directory

- `Get-ADUser -Identity username  # Get user details`
- `Get-ADGroupMember -Identity GroupName  # List group members`
- `Search-ADAccount -LockedOut  # Find locked accounts`
- `Unlock-ADAccount -Identity jdoe  # Unlock account`
- `New-ADUser -Name "John Doe" ...  # Create user`
- `Add-ADGroupMember -Identity Group -Members u  # Add to group`

DNS and Network

- `ipconfig /all  # Full network config`
- `ipconfig /release  # Release IP lease`
- `ipconfig /renew  # Request new IP from DHCP`
- `ipconfig /flushdns  # Clear DNS cache`
- `ping 8.8.8.8  # Test internet routing`
- `ping google.com  # Test DNS resolution`
- `nslookup hostname  # DNS lookup`
- `tracert hostname  # Trace network path`

Group Policy

- `gpupdate /force  # Force immediate GP refresh`
- `gpresult /r  # Show applied GPOs`
- `gpresult /h C:\report.html  # Generate HTML GP report`

Domain Controller Health

- `dcdiag /test:all` # Full DC diagnostic
- `netdom query fsmo` # Show FSMO role holders
- `repadmin /showrepl` # Show AD replication status

File and Share

- `net use H: \\FS01\HR` # Manual drive map
- `net share` # List all shares
- `icacls C:\Shares\HR` # View NTFS permissions

Services

- `services.msc` # Open Services manager
- `net stop spooler` # Stop Print Spooler
- `net start spooler` # Start Print Spooler
- `Get-Service -Name spooler` # Check service status
- `Restart-Service -Name spooler` # Restart service

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12. Complete Lab Status Summary

Component	Status	Notes
VirtualBox Hypervisor	**Complete**	Host machine running, all VMs operational
DC01 — Windows Server 2022	**Complete**	Primary domain controller, fully configured
FS01 — Windows Server 2022	**Complete**	File server, domain joined, shares configured
Workstation VM	**Complete**	Domain joined, end user simulation
pfSense Firewall/Router	**In Progress**	Installed — initial configuration in progress
BitLocker — DC01	**Complete**	C: drive encrypted, XTS-AES, GPO workaround, verified
BitLocker — FS01	**Deferred**	Pending Guest Additions installation
Active Directory	**Complete**	CORP.local domain, OUs, security groups, users
DNS	**Complete**	Authoritative for CORP.local, forwarding configured
DHCP	**Complete**	Scope configured, IP assignment working
Group Policy	**Complete**	Drive mapping GPO deployed and verified
NTFS Permissions	**Complete**	Dept shares with RBAC security group access
Home Directories	**Complete**	\\FS01\HomeDirectories, H: drive mapped
PowerShell — Single User	**Complete**	Interactive provisioning with confirmation gate
PowerShell — Bulk CSV	**Complete**	6/6 Justice League test — all success
Spiceworks Help Desk	**Complete**	Cloud platform, agent on DC01, 4 tickets resolved
Windows Server Backup	**Complete**	Scheduled nightly jobs, verified restore, dedicated E: volume
WSUS	**Complete**	Installed on DC01, GPO deployed, daily sync configured
Azure Entra ID Connect	**Planned**	Hybrid identity sync — needs Azure account
Oracle Cloud Free Tier	**Planned**	Hybrid cloud demo — free tier, no credit card
Duo MFA	**Planned**	Free tier at duo.com
Network Diagram	**Planned**	Draw.io — document full lab topology
GitHub Profile	**Planned**	PowerShell script portfolio — public facing

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